

Combinatorics PhD Exam, May 2023

1. Let $a_1, a_2, \dots, a_{10}$ be positive integers not exceeding 100. Prove that there are disjoint, nonempty subsets S and T of $[10]$ so that

$$\sum_{i \in S} a_i = \sum_{j \in T} a_j.$$

2. Prove that the number of permutations of length $n + 1$ that have exactly two cycles is equal to the number of all cycles in all $n!$ permutations of length n . (For instance, if $n = 3$, then both of these numbers are equal to 11.)
3. Let P be a convex polyhedron whose faces are all either a -gons or b -gons, and whose vertices are each incident to three edges. Let p_a, p_b , and n respectively denote the number of a -gonal faces, b -gonal faces, and vertices of P . Prove that $p_a(6 - a) + p_b(6 - b) = 12$.
4. A city has three high schools, each of them attended by n students. Each student knows exactly $n + 1$ students who attend a high school different from his. Prove that we can choose three students, one from each school, so that each of them knows the other two.
5. Let L_n be the total number of leaves in all binary plane trees with n vertices. These are trees in which all non-leaf vertices have one or two children, and every child is a left child or a right child, even if it is an only child. So $L_1 = 1$, $L_2 = 2$, and $L_3 = 6$. Find an explicit formula for the numbers L_n .
6. Let $c : S \rightarrow \{0, 1\}^*$ be a prefix-free code in which b_i codewords have length i . Prove that $\sum_i \frac{b_i}{2^i} \leq 1$.
7. Let T be a rooted non-plane 2-tree (that is, every non-leaf vertex has two children) with n leaves. For instance, for $n = 4$, there are two such trees, one in which each leaf is at edge-distance two from the root, and one in which the edge-distances of the leaves from the root are 1, 2, 3, and 3. Let $\text{sym}(T)$ be the number of non-leaf vertices v of T such that the two children of v are the roots of identical subtrees. Your answers to parts (a) and (b) can contain $\text{sym}(T)$, but your answer to part (c) should not.
- How many automorphisms does T have?
 - How many different ways are there to bijectively label the leaves of T with the numbers $1, 2, \dots, n$?
 - Find an explicit formula for $\sum_T \frac{1}{|\text{Aut}(T)|}$, where the sum is taken over all rooted non-plane 2-trees with n leaves. For example, for $n = 4$, the two trees mentioned above have, respectively, eight and two automorphisms, so the requested sum is $\frac{1}{8} + \frac{1}{2} = \frac{5}{8}$.
8. Let k be a fixed positive integer, and let $F_k(z) = \sum_{n \geq k} c(n, k) z^n / n!$, where $c(n, k)$ is the signless Stirling number of the first kind.
- Find $F_k(z)$ in an explicit form.
 - Is $F_k(z)$ an algebraic power series over $\mathbb{C}$?
 - Is $F_k(z)$ a d -finite power series over $\mathbb{C}$?